CS425 Group Final Report

Group 4

AI Medical Chatbot

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1. Introduction and Motivation

As mentioned in the project proposal, our team identified access to reliable medical information as a recurring societal need. People are increasingly turning to large language models (LLMs) such as ChatGPT to answer questions on medical symptoms and advice, with 29% of respondents to a study conducted by Yun & Bickmore (2025) doing so. Yun & Bickmore (2025) noted 90% of survey respondents utilised online means, including chatbots, to find information on health advice. Despite the increased accessibility (Clark & Bailey, 2024) to such information through more natural-language, prose-based inputs compared to traditional keyword search, chatbot-based queries are a double-edged sword. The non-deterministic nature of LLM outputs, colloquially known as “hallucinations”, could result in incorrect outputs returned to users. This is highly dangerous to health and safety.

Over the course of this project, our team tuned a small transformer model, Google Flan-T5, on a medical information dataset, and grounded it with search information. We evaluated the performance of the model with the baseline Flan-T5, utilising ROUGE-L and BERTScore

Evaluation was conducted on the following configurations, with the test dataset:

- Base Flan-T5 model
- Base Flan-T5 with Retrieval
- Medical QA-Fine-Tuned Flan-T5*
- Medical QA-Fine-Tuned Flan-T5 with Retrieval System

*Evaluation was conducted on multiple iterations of fine-tuning on different LoRA hyperparameters. The final comparison utilising the configurations from version 9 of the tuning.

The hypothesis was that BERTScore and ROUGE-L, alongside a qualitative analysis of the text, would show that the *Medical QA-Fine-Tuned Flan-T5 with Retrieval System* configuration would generate the best results. This is followed by the other configurations in reverse order (i.e fine-tuning only, base + retrieval and base only). This is as the domain-specific tuning and retrieval was hypothesised to provide greater context and answering capacity for medical questions in the test set.

2. Preparing the data

2.1 Data Source

The primary dataset used for fine-tuning the model was the ruslanmv/ai-medical-chatbot QA Corpus, which is publicly available on HuggingFace. This corpus serves as a robust foundation for medical question-answering tasks, consisting of a CSV file containing approximately 257,000 doctor-patient exchanges.

Link to the dataset: https://huggingface.co/datasets/ruslanmv/ai-medical-chatbot

The raw dataset includes the following three key fields:

- Description (str): The original question or query posed by the patient.
- Patient (str): Additional background, context, or follow-up details provided by the patient.
- Doctor (str): The comprehensive answer or medical advice provided by the doctor.

This structure allows the model to learn from realistic, multi-turn interactions, making it suitable for generating conversational medical responses. An example of the dataset is shown in Figure 1.

Description string · *lengths* 1↔152 99.8%	Patient string · *lengths* 1↔1.78k 99.3%	Doctor string · *lengths* 2↔1.14k 95.1%
Q. What does abutment of the nerve root mean?	Hi doctor,I am just wondering what is abutting and abutment of the nerve root means in a back issue. Please explain. What treatment is required for annular bulging and tear?	Hi. I have gone through your query with diligence and would like you to know that I am here to help you. For further information consult a neurologist online -->

Figure 1: Example of raw dataset

2.2 Preprocessing

To ensure the quality of the training data and optimize the performance of the T5 model, an offline preprocessing pipeline was implemented. The raw dataset underwent several stages of cleaning, filtering, and formatting before being utilized for fine-tuning. More details are provided in section 5.1.

3. Model Architecture & Fine Tuning

3.1 Flan-T5 Model

Further research by the team uncovered that the Google Flan-T5 series of models were much more suitable for the fine-tuning task at hand.

Baseline inference on the Flan-T5 model versus the base T5 showed that the Flan-T5 model returned more coherent outputs even without the fine-tuning. This is likely as the Flan-T5 series has undergone further supervised tuning on “instruction-style datasets”. These are a diverse set of different natural language processing tasks across a variety of domains, such as question answering, summarising and classification, to name a few (Longpre et. al., 2023). Longpre et. al. (2023) also showcase how this diversity in task training types and exposure to different prompting techniques, such as few shot and zero shot enables the resultant Flan-T5 model to better generalise across unseen data by “3-17%+ across [different] evaluation settings”.

The Flan-T5-Base (248M parameters) was selected as a reasonable choice for training, given the limited compute access and time the team had for the project. This is similar to the original T5-base proposed, which has 220M parameters

3.2 LoRA Fine Tuning - Introduction

Fine-tuning using Low-Rank Adaptation (LoRA) was the primary method used for the project. Hu (2021) describes it as a parameter-efficient method inserting and tuning small adapter layers into the model for the tuning proper while keeping the original weights frozen. This approach, which reduces the number of trainable weights to a fraction of the original (for the final model, v9, around 9 million parameters, or under 4% of the original number of weights), thereby making tuning faster and more efficient. Indeed, on the RTX 3090 GPUs on the SCIS cluster, training took a reasonable 2.5 to 4 hours per trial, depending on the hyperparameter configuration.

The fine-tuning code's primary packages were PyTorch, peft and transformers, utilising the latter's T5Seq2SeqTrainer and T5Seq2SeqTrainingArguments classes in particular. Because Flan-T5 is architecturally the same as the base T5, we were able to utilise the existing ecosystem of methods for the fine-tuning process.

Adafactor, being a memory-efficient optimiser with performance comparable to AdamW (Shazeer & Stern, 2018) was selected for use again because of its memory efficiency on our limited compute availability. We were also interested in the adaptive learning rate mechanism it has, which could stabilise training while optimising learning rates. It uses a step size multiplied by the root-mean-square magnitude of its parameters (Shazeer & Stern, 2018). This effectively updates step scales in proportion to parameter magnitude, enabling the desired balance between learning rate optimisation and training stability.

The final iteration of the tuned model targeted both the encoder and decoder attention components as well as feed-forward network weights, with Adafactor as optimiser and its adaptive learning rate functionality enabled. We discuss how the team got to this configuration in the next section.

3.3 LoRA Fine Tuning - Mini-Ablation Studies and Findings

Hyperparameter configuration was a major decision set that had to be made for model performance. Initially, it was thought that decoder attention component tuning, on a fixed baseline learning rate with warmup and gradient clipping for stability would be sufficient for the tuning. However, initial results were quite poor.

The team then went to trial different configurations of hyperparameters, conducting "mini-ablation" studies to see what kind of results would be yielded. Some early runs used incorrect hyperparameter settings, resulting in suboptimal or inconclusive outcomes. Among the major settings that yielded improvements in results were:

- Extending tuning to the encoder layer
- Enabling Adafactor's Adaptive Learning Rate mechanism
- Extending tuning to the feed-forward network

Extending tuning to the encoding layer likely aided performance because of the nature of the dataset and task at hand, which is highly specific in a single domain. Encoder representations could then be refined that would produce numerical representations better

aligned with this more specialised input dataset. This likely aided the effectiveness of generation at the decoder layers.

Kunz (2025), while a relatively new conference paper, conducted ablation studies showing that conducting parameter-efficient fine-tuning on the feed-forward networks of transformer models yielded the best performance of any component. On our model, it did increase scores, but only marginally, and we discuss this briefly later in Section 6.

3.4 Retrieval System with Google Search API

To ground the chatbot's response in reliable medical information, the team built a retrieval system grounded by the Google Search API. On user query, the query would be passed into the API, and the top document result returned, and cleaned of excess HTML. The returned text snippet would be appended behind the user query and then passed to the model for inference.

The team whitelisted the following sources to search medical information from, which were deemed reliable:

- www.cda.gov.sg/
- www.moh.gov.sg/
- medlineplus.gov/
- www.who.int/

4. Key Settings for Hyperparameters

Key LoRA fine-tuning hyperparameters include changes to rank and alpha, enabling of Adafactor's adaptive learning rate. Some baseline settings were inspired from Shojaei (2024)'s tutorial on fine tuning with LoRA.

LoRA hyperparameters

To target both attention and feed-forward networks for tuning, and to use a rank consummate with the size of the dataset (~180K training items)

```
r=16,
lora_alpha=32,
lora_dropout=0.1,
target_modules=r"(encoder|decoder)\..*((SelfAttention|EncDecAttention)\.(q|k|v|o)$|
                (DenseReluDense\.(wi|wo)$))"
```

Adafactor adaptive learning rate hyperparameters

To enable the memory-efficient Adafactor and its adaptive learning rate setting

```
optim="adafactor",
optim_args="clip_threshold=1.0,
                relative_step=True,
                scale_parameter=True,
                warmup_init=True"
```

Evaluation and Save Strategy - *Frequent evaluation of the model to avoid overfitting*

eval_strategy="steps",
eval_steps=1000,
save_strategy="steps",
save_steps=1000,

Warmup Rate and Scheduler
Stabilise early training and improve generalisation
weight_decay=0.01,
warmup_ratio=0.1,
lr_scheduler_type="linear",

Batch Size
For fitting a large effective batch size relative to GPU memory
per_device_train_batch_size=8,
gradient_accumulation_steps=4,
per_device_eval_batch_size=8,

Generation Hyperparameters (for evaluation)
To ensure outputs during evaluation (quantitative and qualitative)
predict_with_generate=True,
logging_steps=50,
generation_max_length=256,
generation_num_beams=4,

Integer Precision and Stability
Improve training reliability (Adafactor has issues with mixed precision)
fp16=False,
label_smoothing_factor=0.1,

Training Hyperparameters
Control the training objective and duration
num_train_epochs=2,
load_best_model_at_end=True,
metric_for_best_model="eval_loss",
greater_is_better=False,

4.1 Changes for the Mini-Ablation Studies

	Major change or setting
v5	rank=16, alpha=32 (baseline)
v8	Enabling dynamic learning rates on Adafactor
v9	**Extending target weights to Feed Forward Network**

5. Code Description

Major components of our code are in the Github repository linked. In particular, we have notebooks for the preprocessing and evaluation steps, and a Python and shell script for training. These are intended to run on the SCIS GPU cluster.

5.1 Preprocessing

Preprocessing first requires the downloading of the dataset from the link in Section 2.1 and the preprocessing is divided into 4 main processes (Data Cleaning, Length Filtering, Conversational Text Processing and Input Formatting and Splitting)

For data cleaning, the initial cleaning phase focused on standardizing the dataset. Approximately 10,300 duplicate entries were identified and removed to ensure unique data points. Following this, text normalization was applied by converting all characters to lowercase, stripping non-ASCII characters to avoid encoding issues, and removing excessive whitespace.

For length filtering, entries were filtered out if

- the question description was fewer than 10 characters,
- the doctor's answer was fewer than 20 characters,
- any field exceeded 1,000 characters.

These constraints successfully removed approximately 24,000 low-quality or overly lengthy rows, resulting in a refined dataset of roughly 222,000 entries.

Also, we processed the conversational text (stripping opening greetings) such as "Hi Doctor," and expressions of gratitude like "Many thanks" from patient inputs. Similarly, formal sign-offs in doctor responses, such as "Regards, Dr. John," were removed to ensure the model predicts concise medical answers rather than generic conversational fillers.

Lastly, the input is formatted into a specific format so that T5 can understand and splitted into training (80%), validation (10%), and testing (10%) sets.

5.2 Model Training

The model was trained on the SMU GPU cluster. The primary script for training is train.py, while the job submission script is batch-v9.sh.

It takes in the inputs from the preprocessing step and converts them into compatible HuggingFace Dataset objects, with tokenisation performed on both question and answers. A Seq2SeqTrainer object, from the HuggingFace Transformers library, is initialised and configured with the relevant hyperparameters as seen earlier in Section 4. The model is then trained on the processed data and saved.

5.3 Quantitative Evaluation

The evaluation notebook (evaluation.ipynb) loads the test dataset and dynamically attaches the appropriate model configuration (either the base FLAN-T5 model alone or with a specified LoRA adapter). For each test sample, the code formats the input into an instruction prompt ("Question: … Answer:"), tokenizes it, and performs autoregressive generation to produce the model's predicted answer. All predictions, ground-truth answers, and questions are stored and exported for inspection.

After inference, the notebook computes quantitative metrics. ROUGE (1, 2, L, Lsum) is calculated using the HuggingFace rouge package to measure lexical overlap between generated outputs and reference answers. Semantic similarity is then assessed with BERTScore, which embeds both texts and evaluates token-level cosine similarity. All metrics are then printed and written to an output text file for record-keeping.

5.4 Inference and Qualitative Evaluation

We conducted a qualitative evaluation on a subset of 21 samples from the test set (excluding index 14 due to the absence of a gold answer). For each selected sample, we generated model outputs under four different configurations: (1) the base FLAN-T5 model without any fine-tuning, (2) the FLAN-T5 model enhanced with LoRA fine-tuning, (3) the base model augmented with Retrieval-Augmented Generation (RAG), and (4) the LoRA-fine-tuned model combined with RAG. This allowed us to compare how fine-tuning and retrieval, individually and together, influenced answer quality, contextual alignment, and factual grounding.

6. Evaluation

6.1 Quantitative Evaluation with ROUGE-L and BERTSore

- **ROUGE-L:** Checks the longest sequence of words that appear in both texts in order.
- **BERTScore:** Evaluates semantic similarity between generated and reference answers using contextual embeddings from BERT.

	ROUGE-L	**BERTScore**
T5-base	0.0773	0.7974
T5-base + LoRA (v5)	0.0812	0.7992
T5-base + LoRA (v8)	0.1165	0.8172
T5-base + LoRA (v9)	0.1186	0.8175
T5-base + LoRA (v9) + RAG	0.1704	0.8495
T5-base + RAG	0.0964	0.8127

The quantitative evaluation using ROUGE-L and BERTScore showed that the fine-tuning and retrieval augmentation improved the performance of the chatbot. The baseline T5-base model had the lowest scores (ROUGE-L = 0.0773, BERTScore = 0.7974), reflecting limited medical domain knowledge.

Introducing LoRA fine-tuning (v5) produced a modest improvement (ROUGE-L = 0.0812, BERTScore = 0.7992), showing that even a simple adapter configuration (rank=16, α=32) helps the model better align its responses with medical ground truths. Enabling dynamic learning rate scheduling via Adafactor in v8 resulted in a more pronounced performance jump (ROUGE-L = 0.1165, BERTScore = 0.8172). This indicates that improved optimisation dynamics helped the model converge to more semantically accurate outputs.

However, extending LoRA to the feed-forward network in v9 yielded only marginal gains over v8 (ROUGE-L from 0.1165 to 0.1186; BERTScore from 0.8172 to 0.8175). This suggests that most influential adjustments were already captured by modifying attention-related parameters, FFN layers contributed limited additional benefit for this task and dataset size.

The most significant improvement was when RAG was introduced. T5-base + LoRA (v9) + RAG achieved the strongest overall performance (ROUGE-L = 0.1704, BERTScore = 0.8495), showing that grounding the model with external medical documents substantially improves both lexical similarity and semantic correctness.

Overall, the results validate that the largest performance gains arise from better optimisation (v8) and retrieval grounding, with FFN adaptation offering diminishing returns.

6.2 Qualitative Evaluation (20 samples)

4 samples were retrieved from each model to evaluate qualitatively. The remaining 16 samples are stored inside the csv files provided in each section.

T5-base

CSV file: medical_qa_t5base_sample_evaluated.csv

```
================ MODEL OUTPUTS (20 rows) ================

=== SAMPLE 1 ===
Question: Medical Question: will multiple blows to the cheekbone result in the cheekbones growing larger?
Patient Info: i d like to direct my question to a bone specialist. this is probably a really silly question however i wanted to ask anyway;
i ve been in a few fights over the last year or so and recently it got me thinking; would taking multiple blows to the cheekbone area result
in my cheekbones growing larger in response to stress? i know it is a bit of an absurd question but i just wanted to ask anyway.
Model Answer: Is stress a factor in the growth of the cheekbones and how can it affect the size of your cheekbone? If so, what is the best c
ourse of action to take in order to achieve this outcome? Is it a good idea to take multiple blows to the cheek bone area after a fight to s
ee if this is the case?

=== SAMPLE 2 ===
Question: Medical Question: q. can headache occur as a side effect of bp medicine?
Patient Info: i am a 50 year old female. i have aches in my limbs and thighs especially at night. it is often characterized by heaviness in
limbs. i am getting headache chiefly in left side of the head. i am taking painkiller at least once in a week for the same. i am taking newt
el 40 mg for blood pressure since a year. kindly help with my condition.
Model Answer: Yes, I have a headache chiefly in left side of the head. I am taking painkiller at least once in a week for the same. i am tak
ing Newtel 40 mg for blood pressure since a year.

=== SAMPLE 3 ===
Question: Medical Question: what could cause mental discomfort after undergoing blood transfusion?
Patient Info: i had a blood transfusion 5 pints and ever since i have no feeling of connections for any family or friends as if they were ne
ver part of my life...and i have no desire to have anything to do with them...is this because i was almost for dead then the 5 pints of bloo
d?
Model Answer: (iii), iv., v. , vii., and viiii. (ivii) , (vv.), viiv. and ixv. (viii).

=== SAMPLE 4 ===
Question: Medical Question: what are the chances of survival if the brain in not functioning?
Patient Info: my grandmother has recently had a triple bypass and has had two grand mal seizures in the past 24 hours since the surgery. the
doctors are now saying that her brain is not responding to her body..what are her chances of survival right now, and why is she having seizu
res? she has never been seizure prone..but is a diabetic.
Model Answer: two grand mal seizures in the past 24 hours since the surgery. the doctors are now saying that her brain is not responding to
her body..what are her chances of survival right now, and why is she having seizures? she has never been seizure prone..but is a diabetic.
```

T5-base + LoRA (v9)

Csv file: medical_qa_t5base+lora_sample_evaluated.csv

```
================ MODEL OUTPUTS (20 rows) ================

=== SAMPLE 1 ===
Question: Medical Question: will multiple blows to the cheekbone result in the cheekbones growing larger?
Patient Info: i d like to direct my question to a bone specialist. this is probably a really silly question however i wanted to ask anyway;
i ve been in a few fights over the last year or so and recently it got me thinking; would taking multiple blows to the cheekbone area result
in my cheekbones growing larger in response to stress? i know it is a bit of an absurd question but i just wanted to ask anyway.
Model Answer: thanks for your question on hcm. i can understand your concern. if the cheekbones are growing larger, then it is likely that y
ou are having a swollen cheekbone. you should consult a bone specialist and get done an x-ray of the cheek and if there is a swelling, then
you can take a topical steroid like acetaminophen. hope i have answered your query, if you have doubt then i will be happy to answer. thanks
for using health care magic. wish you a very good health.

=== SAMPLE 2 ===
Question: Medical Question: q. can headache occur as a side effect of bp medicine?
Patient Info: i am a 50 year old female. i have aches in my limbs and thighs especially at night. it is often characterized by heaviness in
limbs. i am getting headache chiefly in left side of the head. i am taking painkiller at least once in a week for the same. i am taking newt
el 40 mg for blood pressure since a year. kindly help with my condition.
Model Answer: thanks for your question on hcm. i can understand your concern. bp is a medication that is used to treat a variety of conditio
ns. if you are having a headache, you should consult a neurologist for a proper diagnosis and treatment. hope i have answered your query. le
t me know if i could assist you further.

=== SAMPLE 3 ===
Question: Medical Question: what could cause mental discomfort after undergoing blood transfusion?
Patient Info: i had a blood transfusion 5 pints and ever since i have no feeling of connections for any family or friends as if they were ne
ver part of my life...and i have no desire to have anything to do with them...is this because i was almost for dead then the 5 pints of bloo
d?
Model Answer: i have gone through your question. i can understand your concern. if you have a history of blood transfusion, you may be havin
g a blood thrombosis. you may also have an ecg, a mri, and a liver function test. you should consult a psychiatrist to rule out any underlyi
ng cause. hope i has answered your query. let me know if i could assist you further. take care

=== SAMPLE 4 ===
Question: Medical Question: what are the chances of survival if the brain in not functioning?
Patient Info: my grandmother has recently had a triple bypass and has had two grand mal seizures in the past 24 hours since the surgery. the
doctors are now saying that her brain is not responding to her body..what are her chances of survival right now, and why is she having seizu
res? she has never been seizure prone..but is a diabetic.
Model Answer: welcome to healthcaremagic.comi have gone through your query. i can understand your concern. if she is diabetic, she has a cha
nce of survival. so, i suggest you to consult a neurologist and get a neurological evaluation done. hope i have answered your query, if you
have doubt then i will be happy to answer. thanks for using health care magic. wish you a very good health.
```

T5-base + RAG

Csv file: medical_qa_t5base+rag_sample_evaluated.csv

```
================ MODEL OUTPUTS (20 rows) ================

=== SAMPLE 1 ===
Question: Medical Question: will multiple blows to the cheekbone result in the cheekbones growing larger?
Patient Info: i d like to direct my question to a bone specialist. this is probably a really silly question however i wanted to ask anyway;
i ve been in a few fights over the last year or so and recently it got me thinking; would taking multiple blows to the cheekbone area result
in my cheekbones growing larger in response to stress? i know it is a bit of an absurd question but i just wanted to ask anyway.
Model Answer: [Head injury - first aid: MedlinePlus Medical Encyclopedia](https://medlineplus.gov/ency/article/000028.htm): The head may loo
k fine, but problems could result from bleeding or swelling inside the skull. Get medical help right away if the person: Becomes very sleepy
Behaves abnormally, or has speech that does not make sense Develops a severe headache or stiff neck Has a seizure Has pupils (the dark centr
al part of the eye) of unequal sizes Is unable to move all or part of an arm or leg Loses consciousness, even briefly Vomits more than once
Then take the following steps: Check the person's airway, breathing, and circulation. However, contact your provider for medical advice and
watch for symptoms of a head injury, which can show up later.
Sources: Head injury - first aid: MedlinePlus Medical Encyclopedia<https://medlineplus.gov/ency/article/000028.htm>

=== SAMPLE 2 ===
Question: Medical Question: q. can headache occur as a side effect of bp medicine?
Patient Info: i am a 50 year old female. i have aches in my limbs and thighs especially at night. it is often characterized by heaviness in
limbs. i am getting headache chiefly in left side of the head. i am taking painkiller at least once in a week for the same. i am taking newt
el 40 mg for blood pressure since a year. kindly help with my condition.
Model Answer: Semaglutide injection may cause other side effects. If you experience a serious side effect, you or your doctor may send a rep
ort to the Food and Drug Administration's (FDA) MedWatch Adverse Event Reporting program online ( https://www.fda.gov/Safety/MedWatch ) or b
y phone ( 1-800-332-1088 )
Sources: Semaglutide Injection: MedlinePlus Drug Information<https://medlineplus.gov/druginfo/meds/a618008.html>

=== SAMPLE 3 ===
Question: Medical Question: what could cause mental discomfort after undergoing blood transfusion?
Patient Info: i had a blood transfusion 5 pints and ever since i have no feeling of connections for any family or friends as if they were ne
ver part of my life...and i have no desire to have anything to do with them...is this because i was almost for dead then the 5 pints of bloo
d?
Model Answer: a pituitary disease or too much or too little iodine in your diet You are at higher risk for hypothyroidism if you: Are a woma
n Are older than age 60 Have had a thyroid problem before, such as a goiter
Sources: Hypothyroidism | Hashimoto's Disease | MedlinePlus<https://medlineplus.gov/hypothyroidism.html>

=== SAMPLE 4 ===
Question: Medical Question: what are the chances of survival if the brain in not functioning?
Patient Info: my grandmother has recently had a triple bypass and has had two grand mal seizures in the past 24 hours since the surgery. the
doctors are now saying that her brain is not responding to her body..what are her chances of survival right now, and why is she having seizu
res? she has never been seizure prone..but is a diabetic.
Model Answer: [Loss of brain function - liver disease: MedlinePlus Medical ...](https://medlineplus.gov/ency/article/000302.htm): [This is c
alled hepatic encephalopathy (HE)] Once you have liver damage, episodes of worsening brain function may be triggered by: Loss of body fluids
(dehydration) Eating too much protein Low blood potassium or sodium levels Blooding in the intestines, stomach, or food pipe (esophagus) Inf
ections Kidney problems Low oxygen levels in the body Shunt placement or complications Surgery Narcotic pain or sedative medicines Disorders
that can appear similar to HE may include: Alcohol intoxication Alcohol withdrawal Bleeding inside the skull ( subdural hematoma ) Brain dis
order caused by lack of vitamin B1 ( Wernicke-Korsakoff syndrome ) In some cases, HE is a short-term problem that can be corrected. People a
re often not able to care for themselves because of these symptoms.
Sources: Loss of brain function - liver disease: MedlinePlus Medical ...<https://medlineplus.gov/ency/article/000302.htm>
```

T5-base + LoRA (v9) + RAG

Csv file: medical_qa_t5base+lora+rag_sample_evaluated.csv

```
================ MODEL OUTPUTS (20 rows) ================

=== SAMPLE 1 ===
Question: Medical Question: will multiple blows to the cheekbone result in the cheekbones growing larger?
Patient Info: i d like to direct my question to a bone specialist. this is probably a really silly question however i wanted to ask anyway;
i ve been in a few fights over the last year or so and recently it got me thinking; would taking multiple blows to the cheekbone area result
in my cheekbones growing larger in response to stress? i know it is a bit of an absurd question but i just wanted to ask anyway.
Model Answer: thanks for your question on hcm. i can understand your concern. if you have a severe headache or stiff neck, you may need a x
ray to rule out a head injury. a complete x-ray of the head and a CT scan of the skull will help in determining the severity of the injury.
hope i have answered your query. let me know if i could assist you further.
Sources: Head injury - first aid: MedlinePlus Medical Encyclopedia<https://medlineplus.gov/ency/article/000028.htm>

=== SAMPLE 2 ===
Question: Medical Question: q. can headache occur as a side effect of bp medicine?
Patient Info: i am a 50 year old female. i have aches in my limbs and thighs especially at night. it is often characterized by heaviness in
limbs. i am getting headache chiefly in left side of the head. i am taking painkiller at least once in a week for the same. i am taking newt
el 40 mg for blood pressure since a year. kindly help with my condition.
Model Answer: welcome to hcm. i have gone through your query and understand your concern. if you have any other side effects, i would sugges
t you to consult a gynaecologist and get evaluated. you may need to take a course of antihistamines like acetaminophen and ibuprofen. you ca
n also take analgesics like diclofenac and acyclovir. hope i has answered your query. let me know if i can assist you further. take care.
Sources: Semaglutide Injection: MedlinePlus Drug Information<https://medlineplus.gov/druginfo/meds/a618008.html>

=== SAMPLE 3 ===
Question: Medical Question: what could cause mental discomfort after undergoing blood transfusion?
Patient Info: i had a blood transfusion 5 pints and ever since i have no feeling of connections for any family or friends as if they were ne
ver part of my life...and i have no desire to have anything to do with them...is this because i was almost for dead then the 5 pints of bloo
d?
Model Answer: welcome to healthcaremagic.comi have gone through your query. i can understand your concern. if you have any doubts then i wou
ld suggest you to consult a gynaecologist and get evaluated. he will be able to help you. hope i have answered your question. let me know if
i could assist you further.
Sources: MediShield Life Benefits | Ministry of Health<https://www.moh.gov.sg/managing-expenses/schemes-and-subsidies/medishield-life/medish
ield-life-benefits/>

=== SAMPLE 4 ===
Question: Medical Question: what are the chances of survival if the brain in not functioning?
Patient Info: my grandmother has recently had a triple bypass and has had two grand mal seizures in the past 24 hours since the surgery. the
doctors are now saying that her brain is not responding to her body..what are her chances of survival right now, and why is she having seizu
res? she has never been seizure prone..but is a diabetic.
Model Answer: welcome to healthcaremagic.comi have gone through your query. i can understand your concern. if you have LTBI, you may be at h
igh risk for TB infection. you should be vigilant about symptoms and seek prompt medical evaluation if any concerns arise. you can also take
a course of anti-TB medication. hope i have answered your question. let me know if i will be able to assist you further.
Sources: Tuberculosis | Communicable Diseases Agency<https://www.cda.gov.sg/public/diseases/tuberculosis/>
```

6.3 Discussion

The qualitative comparison of the models' outputs are shown below. The comparison of models is explained using the 20 samples in corresponding csv files, rather than just evaluating the 4 samples outputs in the previous section.

T5 Base:
The pre-trained T5 Base model demonstrated a fundamental inability to perform the specific medical QA task without fine-tuning. The model simply paraphrased or repeated the patient's information rather than providing a solution. For instance, in Sample 4 (Seizures), instead of offering a prognosis, the model merely repeated the input phrase "two grand mal seizures in the past 24 hours" and "she has never been seizure prone." Furthermore, when the model could not find a logical continuation, it collapsed into severe hallucinations. This is most evident in Sample 3 (Blood Transfusion), where the model failed to generate any text and instead outputted a nonsensical sequence of Roman numerals "((iii), iv. , v. , vii...)", proving it lacked both the domain knowledge and the structural understanding of the QA format.

T5 Base + LoRA:
From the results, LoRA successfully solved the structural and stylistic issues of the baseline. The model adapted to the specific "doctor" persona present in the training data, consistently initiating responses with empathetic openers like "Welcome to healthcaremagic.com" (Sample 4) or "I can understand your concern." In Sample 19 (ASO Titer), the model generated a fluent, confident response regarding penicillin and joint pain that mimicked human medical advice perfectly. While it sounds authoritative, it offers no evidence or citations to verify its claims, making it prone to "confident hallucinations" where the advice sounds correct but cannot be factually checked by the user, suggesting that RAG integration would help in this case.

T5 Base + RAG, no LoRA:
The Retrieval-Augmented Generation (RAG) feature injected information retrieval into the query. However, this resulted in outputs becoming drier and more definition-heavy rather than a normal doctor-patient dialogue. In Sample 16 (Maple Syrup Urine Disease), the model provided highly specific, accurate medical details, such as quoting "branched-chain alpha-keto acid dehydrogenase complex". Also, for each answer, a "Source" section was provided to cite the information that it retrieved from. However, most of the response lacked the empathetic "bedside manner" of the LoRA model, often dumping complex medical jargon without the comforting introductory phrases required for a user-facing healthcare application.

T5 Base + RAG + LoRA:
The hybrid model achieved the highest qualitative standard by combining the empathetic persona of LoRA with the factual precision of RAG. In Sample 16, unlike the pure RAG model, the hybrid version contextualized the technical definition of Maple Syrup Urine Disease within a supportive sentence structure ("I can understand your concern... Maple syrup urine disease is..."), followed by a specific source citation. Similarly, in Sample 5 (PCOS), the model successfully linked the patient's "bulky ovaries" to Polycystic Ovary Syndrome while maintaining a helpful tone.

7. References

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